

Kalyani Patrikar

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Building **computational pipelines** with quantum chemical methods, physics-based models, or AI models; to uncover mechanisms, enhance system performance, and design new materials

EXPERIENCE

Postdoctoral Research Associate, *University of Liverpool*

PRESENT

- Developed **probabilistic model** for charge transport in mixed conducting polymers with **time variant disorder**
- Developing **physics-based model** for charge transfer at device interfaces

Early Career Fellow, *Indian Institute of Technology Gandhinagar*

Aug 2022 – Oct 2024

One of the highest postdoctoral level fellowships offered by any Indian institute, secured with independent proposal; led publications and mentored students

- Proposed** the **mechanism** for contact resistance in organic devices, demonstrated by combining device characterisation with ab initio computations and **physics-based model**
- Developed a **physics-based model** to determine efficiency of **charge transfer** across metal-organic interfaces based on ab initio computations; applied for organic devices and AuNP catalysts
- Developed **multiscale simulation pipeline** involving **ab initio** computations, semi empirical QM, and **probabilistic model**, mapping molecular structure to device performance, validated against experimental data
- Designed **surrogate** general purpose **graph attention network** for accurate prediction of diverse molecular properties from structure, robust across dataset size
- Developed** automated fragmentation **algorithm** to generate new molecules and analyse with neural network

Senior Research Assistant, *Center for Excellence in Nanoelectronics*

Jul 2013– Dec 2014

- Designed and fabricated oxide nanoparticles embedded organic transistors for toxic gas sensors

SKILLS

ML Frameworks : GNN & GAT, Random Forest, PINN

Modeling : Probabilistic model, physics-based models, ab initio computations

Scientific Software : **CP2K**, **Gaussian16**, Quantum Espresso, xTB, GROMACS, Jmol, VMD

Programming : Python(**PyTorch**, PyG, scikit-learn, RDKit, ASE, MACE, NumPy), Bash (& SLURM)

KEY PUBLICATIONS (COMPLETE LIST AT kalyanipatrikar.github.io/publications)

Journal (*Equal Contribution Authors, † Corresponding Author)

- Kalyani Patrikar, Keval Patadia, Rudranarayan Khatua, Anirban Mondal, "From Molecules to Devices: A Multiscale Approach to Evaluating Organic Photovoltaics" *J. Chem. Theory Comput.*, 20, 22, 2024 [[10.1021/acs.jctc.4c01029](https://doi.org/10.1021/acs.jctc.4c01029)] < **Computational pipeline: molecule in device performance out** >
- Dinesh Trivedi*, Kalyani Patrikar*, Anirban Mondal, "Dataset Size Invariant Graph Attention Network for Prediction of Diverse Molecular Properties", *Mol. Syst. Des. Eng.*, 9, 2024 [[10.1039/D4ME00113C](https://doi.org/10.1039/D4ME00113C)] < **Surrogate GAT model, accurate for electronic as well as optical molecular properties** >
- Bibhas Das, Kalyani Patrikar, Atharva Keny, Anirban Mondal, "From fragments to function: data-driven design of high-performance non-fullerene acceptors for organic photovoltaics" *Mol. Syst. Des. Eng.*, 11, 269-291, 2026 [[10.1039/D5ME00182J](https://doi.org/10.1039/D5ME00182J)] < **Automated fragmentation and combinatorial generation of novel molecules** >

4. Kalyani Patrikar[†], Valipe Ramgopal Rao, Dinesh Kabra, Anirban Mondal[†], "Understanding the Microscopic Origin of the Contact Resistance at the Polymer–Electrode Interface", *ACS Appl. Mater. Interfaces*, 15 (42), 49427, 2023 [10.1021/acsami.3c10260] < **Mechanism of interface charge transfer in organic transistors** >
5. Kalyani Patrikar[†], Anirban Mondal[†], "Deciphering the Microscopic Phenomenon Behind Contact Resistances in Interlayer Functionalized Electrodes and Organic Semiconductors", *Physical Review Materials*, 8, 054606, 2024 [10.1103/PhysRevMaterials.8.054606] < **Mechanism of charge transfer at functionalised interfaces** >

Conference (*Presenting Author)

1. K. Patrikar*, C. Burke, A. Troisi, "Charge Transport Enhancement with Dynamic Disorder", Oral presentation at *European Conference in Molecular Electronics 2025* < **Probabilistic model for charge transport in dynamically disordered systems** > Awarded **Best Contributed Talk**

EDUCATION

Doctor of Philosophy

Indian Institute of Technology Bombay

Electrical Engineering

Jan 2015 – Dec 2021

Master of Technology

Indian Institute of Technology Bombay

Materials Science

Jul 2011 – Jun 2013

Bachelor of Technology

Visvesvaraya National Institute of Technology

Metallurgical & Materials Engineering

Jul 2007 – May 2011

PHD THESIS

Bulk and Interfacial Charge Transport in Organic Field Effect Transistors

- Demonstrated **novel** charge doping **mechanism** of self-assembled monolayers at transistor contacts; developed **device model** based on ab initio computations and device characterisation
- Established new molecular design paradigm for order-of-magnitude **device performance** improvement based on device characterisation, ab initio computations, and **polaron transport** analysis
- Developed **multiscale model** simulating measured cryogenic device characteristics of organic transistors

TEACHING, MENTORSHIP, COMMUNICATION

- **Teaching:** Flexible Electronics (2 Credit Course) *IIT Gandhinagar*
- **Demonstrator:** Introduction to Spectroscopy, **Chemical Data Discovery and Design** *University of Liverpool*
- **Teaching Assistant:** Physics of Transistors, **Machine Learning**, Microelectronics Lab, Communication Skills, VLSI Technology, Thermodynamics *IIT Bombay*
- **Students Mentored:** B.Tech and MSc. students for thesis project work *IIT Gandhinagar*

EXTRACURRICULAR ACHIEVEMENTS

- Organised and anchored events as the Institute Executive Member in IIT Bombay Academic Council
- Completed high altitude treks in Himalayas, India